

FEATURES

- 512K x 8 organization
- Single +5V power supply
- +12.5V programming voltage
- Fast access time: 120/150 ns
- Totally static operation

- Completely TTL compatible
- Operating current: 60mA
- Standby current: 100μA
- Package type:
 - 32 pin ceramic DIP, plastic DIP

GENERAL DESCRIPTION

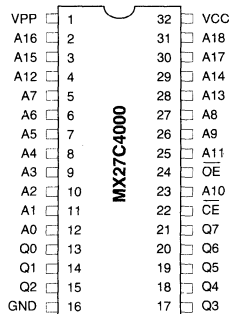
The MX27C4000 is a 5V only, 4M-bit, ultraviolet Erasable Programmable Read Only Memory. It is organized as 512K words by 8 bits per word, operates from a single +5 volt supply, has a static standby mode, and features fast single address location programming. All programming signals are TTL levels, requiring a single pulse. For

programming outside from the system, existing EPROM programmers may be used. The MX27C4000 supports a intelligent quick pulse programming algorithm which can result in programming times of less than two minutes.

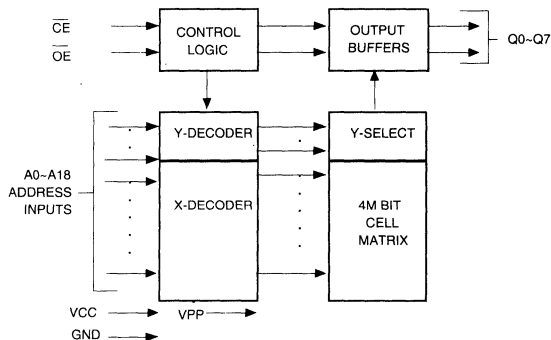
This EPROM is packaged in industry standard 32 pin dual-in-line packages.

PIN CONFIGURATIONS

32 CDIP/PDIP



BLOCK DIAGRAM



PIN DESCRIPTION

SYMBOL	PIN NAME
A0-A18	Address Input
Q0-Q7	Data Input/Output
$\overline{CE}$	Chip Enable Input
$\overline{OE}$	Output Enable Input
VPP	Program Supply Voltage
VCC	Power Supply Pin (+5V)
GND	Ground Pin

FUNCTIONAL DESCRIPTION

THE ERASURE OF THE MX27C4000

The MX27C4000 is erased by exposing the chip to an ultraviolet light source. A dosage of 15 W seconds/cm² is required to completely erase a MX27C4000. This dosage can be obtained by exposure to an ultraviolet lamp — wavelength of 2537 Angstroms (Å) — with intensity of 12,000 μW/cm² for 15 to 20 minutes. The MX27C4000 should be directly under and about one inch from the source and all filters should be removed from the UV light source prior to erasure.

It is important to note that the MX27C4000, and similar devices, will be cleared for all bits of their programmed states with light sources having wavelengths shorter than 4000 Å. Although erasure times will be much longer than that with UV sources at 2537Å, nevertheless the exposure to fluorescent light and sunlight will eventually erase the MX27C4000 and exposure to them should be prevented to realize maximum system reliability. If used in such an environment, the package window should be covered by an opaque label or substance.

THE PROGRAMMING OF THE MX27C4000

When the MX27C4000 is delivered, or it is erased, the chip has all 4M bits in the "ONE", or HIGH state. "ZEROS" are loaded into the MX27C4000 through the procedure of programming.

The programming mode is entered when 12.5 ± 0.5 V is applied to the VPP pin, OE is at VIH, and CE is at VIL.

For programming, the data to be programmed is applied with 8 bits in parallel to the data pins.

The flowchart in Figure 1 shows MXIC's interactive algorithm. Interactive algorithm reduces programming time by using short programming pulses and giving each address only as many pulses as is necessary in order to reliably program the data. After each pulse is applied to a given address, the data in that address is verified. If the data is not verified, additional pulses are given until it is verified or the maximum is reached. This process is repeated while sequencing through each address of the MX27C4000. This part of the algorithm is done at VCC = 6.0V to assure that each EPROM bit is programmed to a sufficiently high threshold voltage. After the interactive programming is completed, an overprogram pulse is given to each memory location; this ensures that all bits have sufficient margin. After the final address is

completed, the entire EPROM memory is verified at VCC = $5V \pm 10\%$.

FAST PROGRAMMING

The device is set up in the fast programming mode when the programming voltage VPP = 12.75V is applied, with VCC = 6.25 V and OE = VIH (Algorithm is shown in Figure 2). The programming is achieved by applying a single TTL low level 100μs pulse to the $\overline{CE}$ input after addresses and data line are stable. If the data is not verified, an additional pulse is applied for a maximum of 25 pulses. This process is repeated while sequencing through each address of the device. When the programming mode is completed, the data in all address is verified at VCC = VPP = $5V \pm 10\%$.

PROGRAM INHIBIT MODE

Programming of multiple MX27C4000s in parallel with different data is also easily accomplished by using the Program Inhibit Mode. Except for CE and OE, all like inputs of the parallel MX27C4000 may be common. A TTL low-level program pulse applied to an MX27C4000 $\overline{CE}$ input with VPP = 12.5 ± 0.5 V and CE LOW will program that MX27C4000. A high-level $\overline{CE}$ input inhibits the other MX27C4000s from being programmed.

PROGRAM VERIFY MODE

Verification should be performed on the programmed bits to determine that they were correctly programmed. The verification should be performed with OE at VIL and CE, at VIH, and VPP at its programming voltage.

AUTO IDENTIFY MODE

The auto identify mode allows the reading out of a binary code from an EPROM that will identify its manufacturer and device type. This mode is intended for use by programming equipment for the purpose of automatically matching the device to be programmed with its corresponding programming algorithm. This mode is functional in the $25^{\circ}\text{C} \pm 5^{\circ}\text{C}$ ambient temperature range that is required when programming the MX27C4000.

To activate this mode, the programming equipment must force 12.0 ± 0.5 V on address line A9 of the device. Two identifier bytes may then be sequenced from the device outputs by toggling address line A0 from VIL to VIH. All other address lines must be held at VIL during auto identify mode.

Byte 0 (A0 = VIL) represents the manufacturer code, and

byte 1 (A0 = VIH), the device identifier code. For the MX27C4000, these two identifier bytes are given in the Mode Select Table. All identifiers for manufacturer and device codes will possess odd parity, with the MSB (DQ7) defined as the parity bit.

READ MODE

The MX27C4000 has two control functions, both of which must be logically satisfied in order to obtain data at the outputs. Chip Enable ($\overline{CE}$) is the power control and should be used for device selection. Output Enable ($\overline{OE}$) is the output control and should be used to gate data to the output pins, independent of device selection. Assuming that addresses are stable, address access time (tACC) is equal to the delay from $\overline{CE}$ to output (tCE). Data is available at the outputs tOE after the falling edge of $\overline{OE}$'s, assuming that $\overline{CE}$ has been LOW and addresses have been stable for at least tACC - tOE.

STANDBY MODE

The MX27C4000 has a CMOS standby mode which reduces the maximum VCC current to 100 μ A. It is placed in CMOS standby when $\overline{CE}$ is at VCC $\pm$ 0.3 V. The MX27C4000 also has a TTL-standby mode which reduces the maximum VCC current to 1.5 mA. It is placed in TTL-standby when $\overline{CE}$ is at VIH. When in standby mode, the outputs are in a high-impedance state, independent of the $\overline{OE}$ input.

TWO-LINE OUTPUT CONTROL FUNCTION

MODE SELECT TABLE

MODE	PINS					
	$\overline{CE}$	$\overline{OE}$	A0	A9	VPP	OUTPUTS
Read	VIL	VIL	X	X	VCC	DOUT
Output Disable	VIL	VIH	X	X	VCC	High Z
Standby (TTL)	VIH	X	X	X	VCC	High Z
Standby (CMOS)	VCC $\pm$ 0.3V	X	X	X	VCC	High Z
Program	VIL	VIH	X	X	VPP	DIN
Program Verify	VIH	VIL	X	X	VPP	DOUT
Program Inhibit	VIH	X	X	X	VPP	High Z
Manufacturer Code	VIL	VIL	VIL	VH	VCC	C2H
Device Code	VIL	VIL	VIH	VH	VCC	40H

NOTES: 1. VH = 12.0 V $\pm$ 0.5 V

2. X = Either VIH or VIL (For auto select)

To accommodate multiple memory connections, a two-line control function is provided to allow for:

1. Low memory power dissipation,
2. Assurance that output bus contention will not occur.

It is recommended that $\overline{CE}$ be decoded and used as the primary device-selecting function, while $\overline{OE}$ be made a common connection to all devices in the array and connected to the READ line from the system control bus. This assures that all deselected memory devices are in their low-power standby mode and that the output pins are only active when data is desired from a particular memory device.

SYSTEM CONSIDERATIONS

During the switch between active and standby conditions, transient current peaks are produced on the rising and falling edges of Chip Enable. The magnitude of these transient current peaks is dependent on the output capacitance loading of the device. At a minimum, a 0.1 μ F ceramic capacitor (high frequency, low inherent inductance) should be used on each device between VCC and GND to minimize transient effects. In addition, to overcome the voltage drop caused by the inductive effects of the printed circuit board traces on EPROM arrays, a 4.7 μ F bulk electrolytic capacitor should be used between VCC and GND for each eight devices. The location of the capacitor should be close to where the power supply is connected to the array.

FIGURE 1. INTERACTIVE PROGRAMMING FLOW CHART

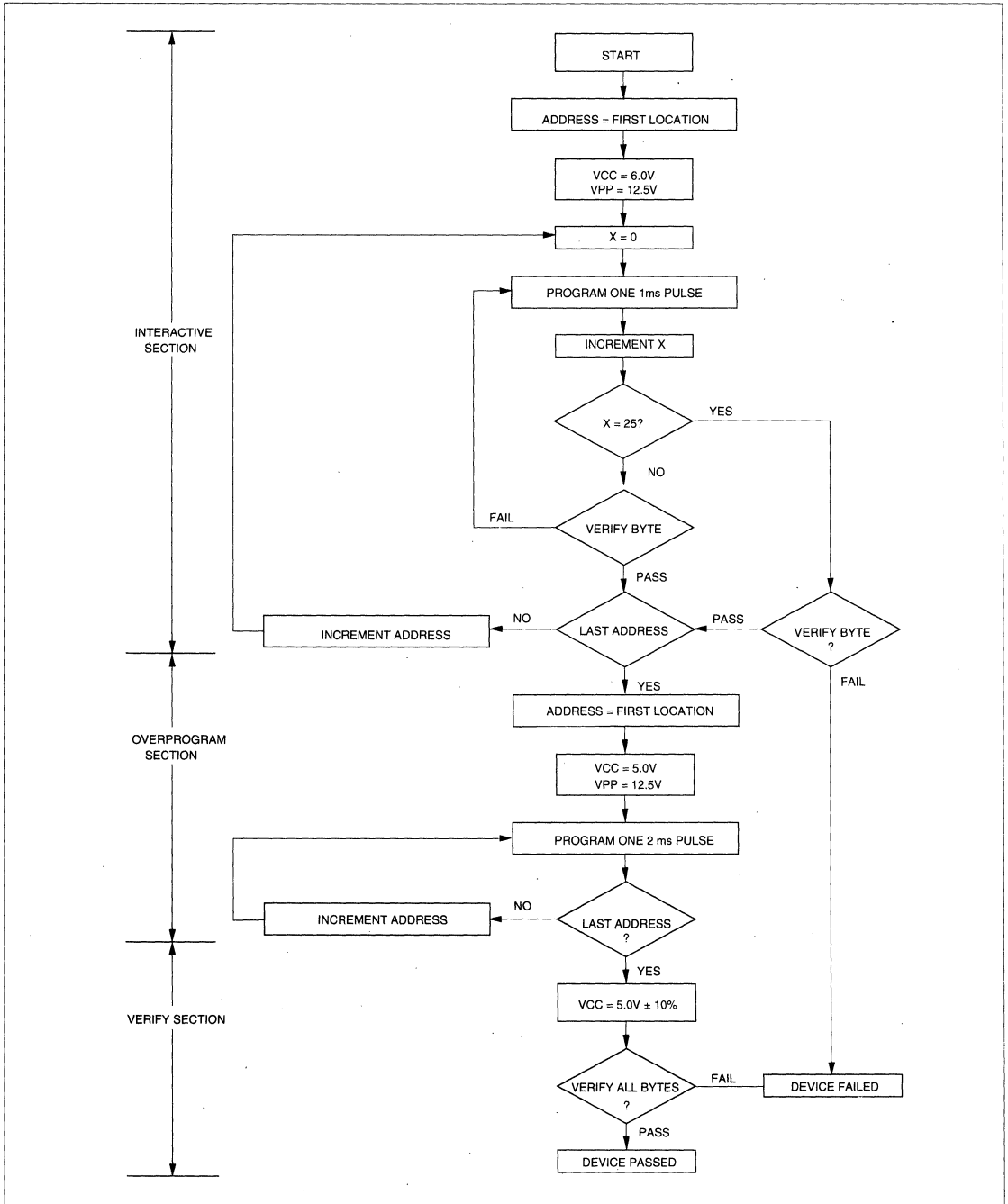
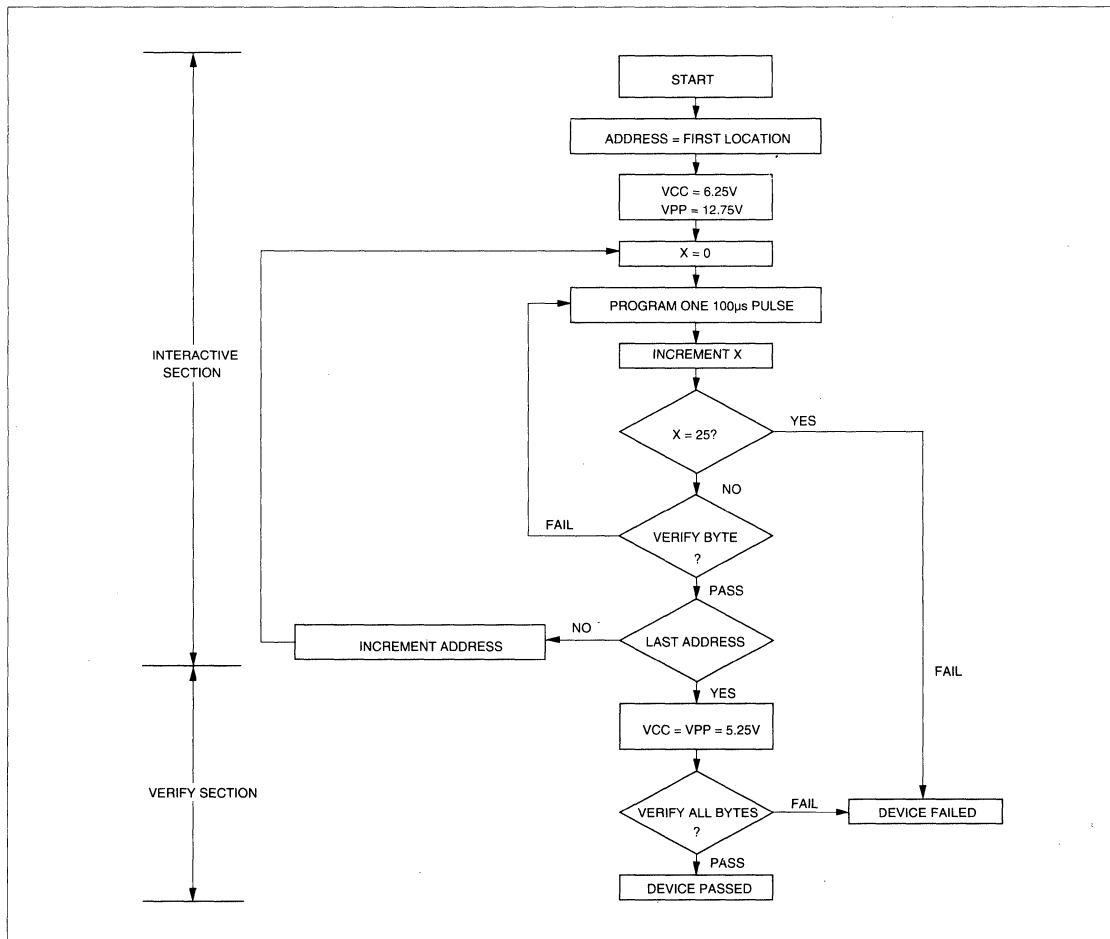
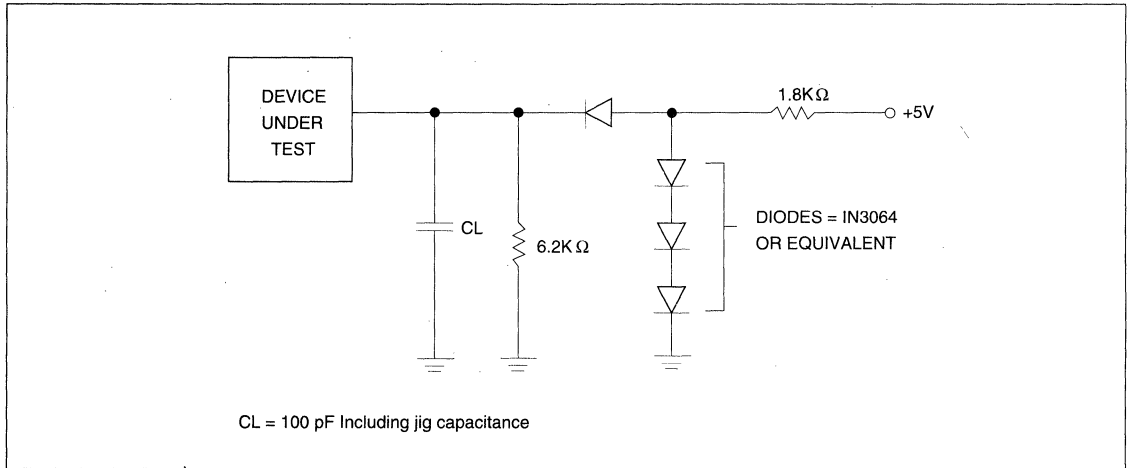


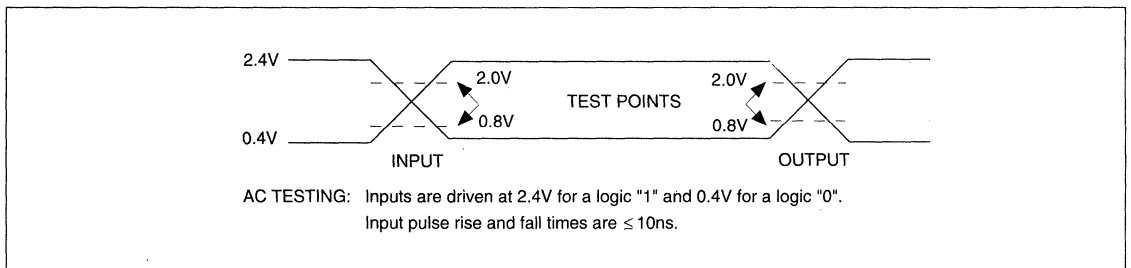
FIGURE 2. FAST PROGRAMMING FLOW CHART



SWITCHING TEST CIRCUITS



SWITCHING TEST WAVEFORMS



ABSOLUTE MAXIMUM RATINGS

RATING	VALUE
Ambient Operating Temperature	0°C to 70°C
Storage Temperature	-65°C to 125°C
Applied Input Voltage	-0.5V to 7.0V
Applied Output Voltage	-0.5V to VCC + 0.5V
VCC to Ground Potential	-0.5V to 7.0V
V9 & VPP	-0.5V to 13.5V

NOTICE:

Stresses greater than those listed under ABSOLUTE MAXIMUM RATINGS may cause permanent damage to the device. This is a stress rating only and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended period may affect reliability.

NOTICE:

Specifications contained within the following tables are subject to change.

DC CHARACTERISTICS TA = 0°C to 70°C, VCC = 5V ± 10%

SYMBOL	PARAMETER	MIN.	MAX.	UNIT	CONDITIONS
VOH	Output High Voltage	2.4		V	IOH = -0.4mA
VOL	Output Low Voltage		0.4	V	IOL = 2.1mA
VIH	Input High Voltage	2.0	VCC + 0.5	V	
VIL	Input Low Voltage	-0.3	0.8	V	
ILI	Input Leakage Current	-10	10	μA	VIN = 0 to 5.5V
ILO	Output Leakage Current	-10	10	μA	VOUT = 0 to 5.5V
ICC3	VCC Power-Down Current		100	μA	$\overline{CE} = VCC \pm 0.3V$
ICC2	VCC Standby Current		1.5	mA	$\overline{CE} = VIH$
ICC1	VCC Active Current		60	mA	$\overline{CE} = VIL, f=5MHz, I_{out} = 0mA$
IPP	VPP Supply Current Read		100	μA	$\overline{CE} = \overline{OE} = VIL, VPP = 5.5V$
IPP2	VPP Supply Current (Program)		50	mA	

CAPACITANCE TA = 25°C, f = 1.0 MHz (Sampled only)

SYMBOL	PARAMETER	TYP.	MAX.	UNIT	CONDITIONS
CIN	Input Capacitance	8	12	pF	VIN = 0V
COUT	Output Capacitance	8	12	pF	VOUT = 0V
CVPP	VPP Capacitance	18	25	pF	VPP = 0V

AC CHARACTERISTICS TA = 0°C to 70°C, VCC = 5V ± 10%

SYMBOL	PARAMETER	27C4000-12		27C4000-15		UNIT	CONDITIONS
		MIN.	MAX.	MIN.	MAX.		
tACC	Address to Output Delay		120		150	ns	$\overline{CE} = \overline{OE} = VIL$
tCE	Chip Enable to Output Delay		120		150	ns	$\overline{OE} = VIL$
tOE	Output Enable to Output Delay		50		65	ns	$\overline{CE} = VIL$
tDF	$\overline{OE}$ High to Output Float, or $\overline{CE}$ High to Output Float	0	35	0	50	ns	
tOH	Output Hold from Address, $\overline{CE}$ or $\overline{OE}$ which ever occurred first	0		0		ns	

DC PROGRAMMING CHARACTERISTICS TA = 25°C ± 5°C

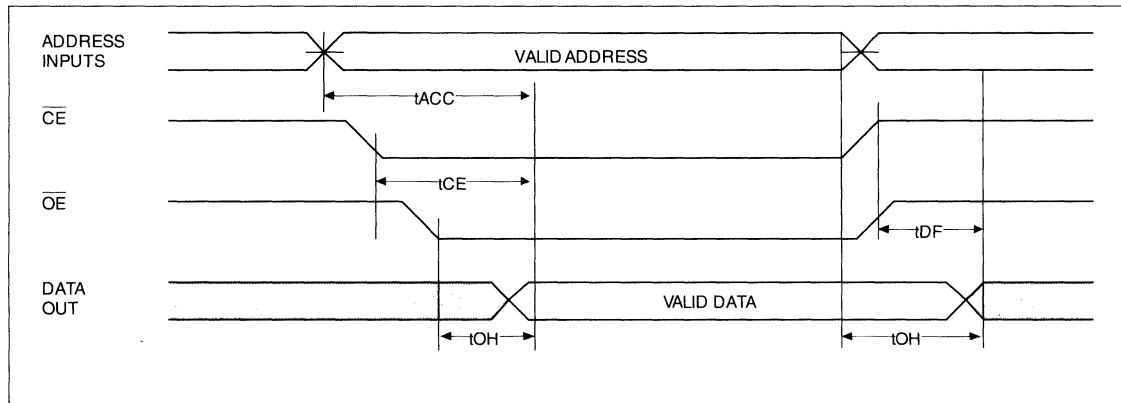
SYMBOL	PARAMETER	MIN.	MAX.	UNIT	CONDITIONS
VOH	Output High Voltage	2.4		V	IOH = -0.40mA
VOL	Output Low Voltage		0.4	V	IOL = 2.1mA
VIH	Input High Voltage	2.0	VCC + 0.5	V	
VIL	Input Low Voltage	-0.3	0.8	V	
ILI	Input Leakage Current	-10	10	μA	VIN = 0 to 5.5V
VH	A9 Auto Select Voltage	11.5	12.5	V	
ICC3	VCC Supply Current (Program & Verify)		50	mA	
IPP2	VPP Supply Current(Program)		30	mA	CE = VIL, OE = VIH
VCC1	Interactive Supply Voltage	5.75	6.25	V	
VPP1	Interactive Programming Voltage	12.0	13.0	V	
VCC2	Fast Programming Supply Voltage	6.00	6.50	V	
VPP2	Fast Programming Voltage	12.5	13.0	V	

AC PROGRAMMING CHARACTERISTICS TA = 25°C ± 5°C

SYMBOL	PARAMETER	MIN.	MAX.	UNIT	CONDITIONS
tAS	Address Setup Time	2.0		μS	
tOES	OE Setup Time	2.0		μS	
tDS	Data Setup Time	2.0		μS	
tAH	Address Hold Time	0		μS	
tDH	Data Hold Time	2.0		μS	
tDFP	CE to Output Float Delay	0	130	nS	
tVPS	VPP Setup Time	2.0		μS	
tPW	CE Program Pulse Width	<i>Fast</i>	95	105	μS
		<i>Interactive</i>	0.95	1.05	mS
tOPW	CE Overprogram Pulse(Interactive)	1.95	2.05	mS	
tVCS	VCC Setup Time	2.0		μS	
tDV	Data Valid from CE		250	nS	
tCES	CE Setup Time	2.0		μS	
tOE	Data valid from OE		150	nS	

WAVEFORMS

READ CYCLE



INTERACTIVE PROGRAMMING ALGORITHM WAVEFORMS(NOTE 1 & 2)

